
Chapter 9: Symmetry

Worksheet 9: Line Symmetry and Rotational Symmetry

Learning Objective: Students will identify lines of symmetry in 2D figures and letters; determine the order of rotational symmetry; distinguish between line symmetry and rotational symmetry; and explore symmetry in nature and architecture.

Worked Example

A square has **4 lines of symmetry** (2 through midpoints of opposite sides, 2 through opposite corners) and **rotational symmetry of order 4** (it maps onto itself at 90, 180, 270, 360).

Questions 1–5: Easy / Recall

Q1. How many lines of symmetry does an equilateral triangle have?

Q2. What is the order of rotational symmetry of a square?

Q3. Does the letter **H** have line symmetry? How many lines?

Q4. What is the angle of rotation for a shape with order-3 rotational symmetry?

Q5. Which of these has rotational symmetry but no line symmetry: S, N, Z?

Questions 6–10: Basic Application

- Q6.** List the number of lines of symmetry for: circle, rectangle, regular pentagon, regular hexagon.
- Q7.** A regular polygon has 8 lines of symmetry. How many sides does it have? What is the order of its rotational symmetry?
- Q8.** Does the number 8 have any symmetry? Describe it.
- Q9.** An equilateral triangle is rotated 120° . Does it look the same? What is its order of rotational symmetry?
- Q10.** Identify two Indian monuments that show line symmetry.

Questions 11–15: Practice and Skill Building

- Q11.** Complete: A regular polygon with n sides has _____ lines of symmetry and rotational symmetry of order _____.
- Q12.** The letter **M** has vertical line symmetry. Write four other capital letters with only vertical line symmetry.
- Q13.** Does a parallelogram (non-rectangular) have any line of symmetry? Does it have rotational symmetry? Explain.

Q14. A snowflake has 6-fold rotational symmetry. At what angles does it map onto itself?

Q15. Draw a rangoli pattern with 4-fold symmetry. (Teacher Verification Required)

Questions 16–18: Higher Order Thinking

Q16. A figure has both line symmetry and rotational symmetry of order 4. Describe all possible angles of rotational symmetry and all lines of symmetry.

Q17. Can a shape have rotational symmetry but no line symmetry? Give an example and explain.

Q18. The capital letters with rotational symmetry of order 2 include H, I, N, S, X, Z. Verify at least 3 of these by rotating 180 and checking.

Questions 19–20: Real-Life Applications

Q19. Look at a 500-rupee note or a coin. Does it have any symmetry? Describe the line(s) of symmetry if any.

- Q20.** A ceiling fan has 3 blades equally spaced. What is the order of rotational symmetry? At what angles does the fan look the same? If the fan has 4 blades, how do the answers change?

Reflection Question: Is there a shape that has an infinite order of rotational symmetry? If so, name it and explain.